

CS 2XY3 Final Exam (Solutions)

Question 1 Solutions - Deadlocks

(a) Four necessary conditions: (1) Mutual exclusion - at least one resource held in non-sharable mode. (2) Hold and wait - a process holds resources while waiting for others. (3) No preemption - resources cannot be forcibly taken. (4) Circular wait - a circular chain of processes each waiting for a resource held by the next.

(b) Yes, there is a deadlock. Cycle: $P1 \rightarrow R2 \rightarrow P2 \rightarrow R3 \rightarrow P3 \rightarrow R1 \rightarrow P1$. P4 is not part of the cycle but is blocked waiting for R2 (held by P2 which is in the cycle).

(c) Banker's Algorithm: requires (1) Max matrix - maximum demand of each process for each resource type. (2) Allocation matrix - current allocation. (3) Available vector - currently free instances of each resource. (4) Need matrix = Max - Allocation. The algorithm checks if granting a request leads to a safe state (a sequence exists where all processes can finish).

Question 2 Solutions - File Systems

(a) Contiguous: Adv - fast sequential/random access. Disadv - external fragmentation, file growth is difficult. Linked: Adv - no external fragmentation, easy growth. Disadv - slow random access (must traverse list), pointer overhead. Indexed: Adv - fast random access, no external fragmentation. Disadv - index block overhead, limited max file size with single index.

(b) Direct: $12 \times 4 \text{ KB} = 48 \text{ KB}$. Single-indirect: $1024 \times 4 \text{ KB} = 4 \text{ MB}$. Double-indirect: $1024 \times 1024 \times 4 \text{ KB} = 4 \text{ GB}$. Triple-indirect: $1024^3 \times 4 \text{ KB} = 4 \text{ TB}$. Max file = $48 \text{ KB} + 4 \text{ MB} + 4 \text{ GB} + 4 \text{ TB}$ (approximately 4 TB).

Question 3 Solutions - Virtual Memory

(a) Page fault process: (1) Trap to OS. (2) Save user registers/state. (3) Check if reference is valid. (4) Find a free frame (or select victim via replacement algorithm). (5) If victim is dirty, write it to disk. (6) Read desired page from disk into frame. (7) Update page table. (8) Restart the instruction.

(b) Reference string 7,0,1,2,0,3,0,4,2,3,0,3,2 with 3 frames:

FIFO: 10 page faults.

LRU: 10 page faults.

(Detailed trace available upon request.)

(c) Thrashing: a process spends more time paging (swapping pages in/out) than executing. Caused by: too many processes competing for limited frames, so each process has fewer frames than its working set requires. Each page-in causes another page-out of a page that will be needed soon.